

NOKIA SPONSERED · OPERATIONALISED BY TSSC · DELIVERED BY MSM TECHNOLOGIES

Advanced AI & ML

Telecom Data Analytics

Day 1 · Module 1

Telecom & 5G Foundations for AI

10-day programme · 4 hours per day · 40 hours

Facilitated by: Ishan Marwah | MSM Technologies

The Nokia logo, consisting of the word "NOKIA" in a blue, sans-serif font.

WhatsApp Group

<https://chat.whatsapp.com/KHwYsU2Eq8r9RE2QiZs3Tn?s=cl&p=a&ilr=1>

- WhatsApp Group:
 - With Your Selfie
 - Name
 - College
 - Semester and Branch



Connectivity and Common Folder

- WiFi Password:
 - SSID : TSSCNoida
 - PWD : Tsscnoida@123

- Google Drive:

<https://drive.google.com/drive/folders/1GcWHVQHu6luYOWKs2Gd-mQXxe2ZnAKU9?usp=sharing>

For the Allocated group to you – please add the name of all the members in your group

 Group 1 - Roshit Ratnadeep , Sahil Sahney, Alavala Bhav...

 Group 2 - Shatakshi, Anamika, Amisha, Yash,Vibhuti

 Group 3 Abhishek , Akanksha, Aryan, Yash,Ansh,Adarsh

Software	Purpose	Notes
MySQL Server + Workbench	Database practice	Full edition; username and password to be retained by the learner
Anaconda (Python + Jupyter)	Development environment	Distribution installer; ML libraries installed during the course
Power BI Desktop	Visualisation and NLQ	Free desktop edition
Git client	Version control and portfolio	Plus a GitHub account created before Day 1
CSV editor (Excel / Notepad++)	Data inspection	Either is acceptable
Grafana OSS	Live KPI monitoring and alerting (Day 7)	Free, self-hosted, AGPLv3; Windows installer
Web browser	Cloud consoles and AI Studio access	Edge or Chrome
Google account	Free Gemini API key for Day 8	Learners almost always have one already
Hugging Face account	Model deployment on Day 9	Free; email address only, no payment details

<https://drive.google.com/drive/folders/1aUCQ17S8b3F8-tG1thWp3MTQJSJ0Bu7Hd?usp=sharing>

NOTEBOOK TO USE

Jupyter Notebook Anaconda – Offline

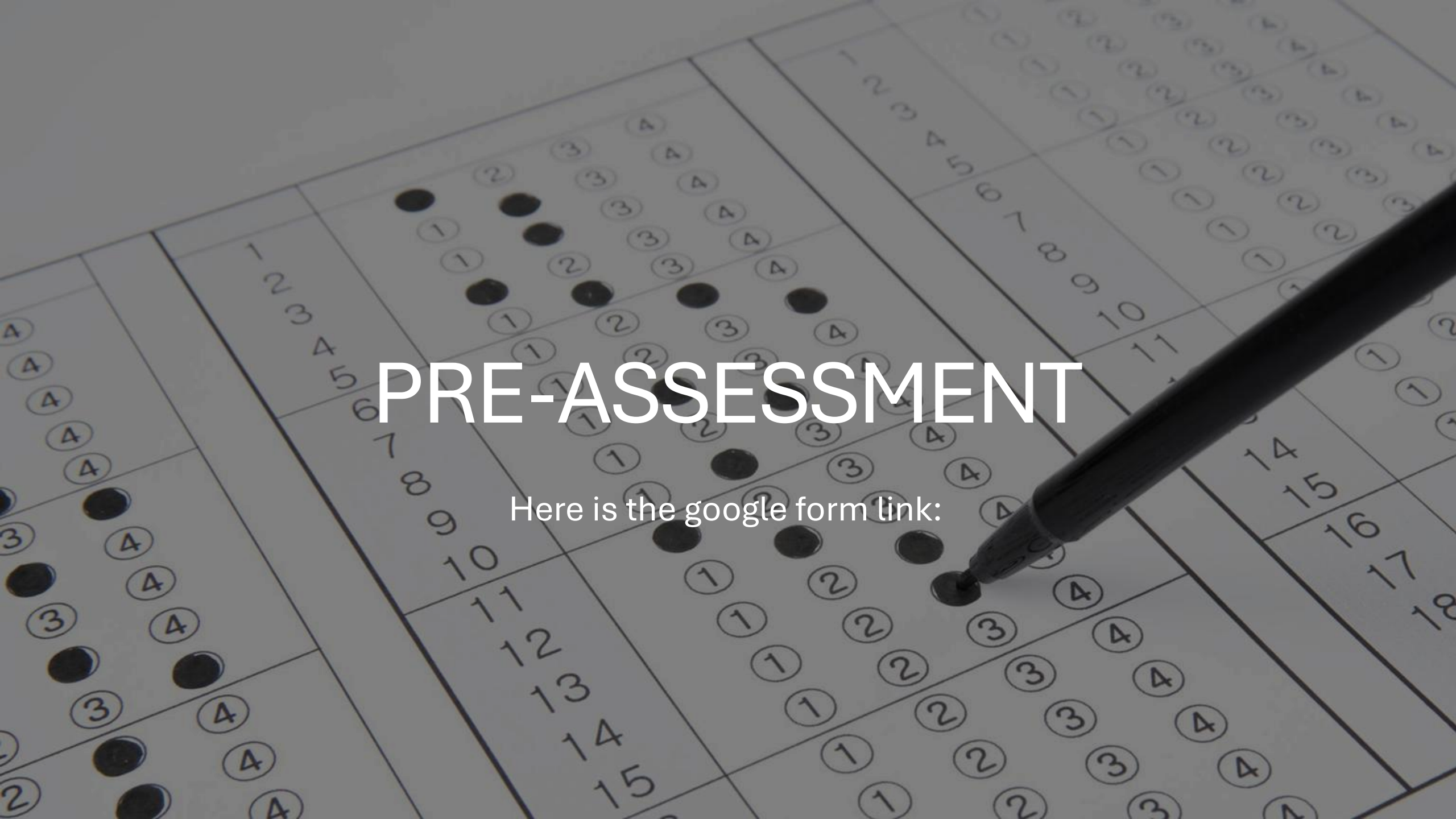
- Pros
 - Full control over environments
 - Works offline and integrates with local files, large datasets
 - Easier installation of geospatial stack (fewer surprises).
 - Better for long-running experiments and production-ish workflows.
- Cons
 - Requires installing Anaconda
 - No free GPU by default (need local compute)

Google Colab – cloud based

- Pros
 - No install - start immediately in a browser.
 - Free access to CPUs, occasional GPUs/TPUs (good for ML practice).
 - Built-in Google Drive integration for storage.
 - Easy to share notebooks (link or GitHub).
- Cons
 - Limited session time (disconnects after inactivity)
 - Privacy: code/data stored in Google Drive or Colab session.

PRE-ASSESSMENT

Here is the google form link:





ISHAN MARWAH

**Senior AI/ML Practitioner ,
O&M Expert &
Enterprise Trainer**

Founder, MSM Technologies
Ex-Qualcomm

Profile Overview

Delivered 1,500+ training man-days globally and led end-to-end transformation for Tier 1 Telcos across the Middle East, APAC, and India.

Certified Generative AI & Agentic AI Practitioner and Enterprise Trainer with 17+ years of cross-industry technical depth.

1,500+

Training Man-Days

50+

Countries

17+ years

Expert in Telecom,

Qualifications

B.Tech – Electronics & Communication · NSDC-Accredited AI Generalist Accelerator ·
Certified Generative AI Practitioner

Achievements

O&M and AI/ML deployments for Telecom, Utilities, Oil & Gas, Energy and enterprise clients. Specializing in connectivity designing , planning , optimization , monitoring solutions, field Ops

[Ishan Marwah](#) | [LinkedIn](#) , Email : Ishan.Marwah@gmail.com

WELCOME

You already know how to build a model. Today you learn what the numbers mean.

Every dataset in this programme comes out of a live mobile network. A column called prb_util, a spike in dropped calls at 21:00, a region whose ARPU is eight rupees below the others — each of these has a physical cause.

An analyst who cannot name that cause builds models that are technically correct and commercially useless. Day 1 exists to make sure that never describes you.

The gap this day closes

- 1 Fit a classifier and report accuracy**
Say what a PM counter physically measures
- 2 Write a clean Pandas transformation**
Explain why a 700 MHz cell behaves unlike a 3.5 GHz cell
- 3 Draw a correlation heatmap**
Say why a 2-point ARPU move is a board-level number

“Domain before data. Data before models. Models before deployment.”

How today runs

Time	Slot	What happens
09:00 – 10:15	Orientation	Programme map, ground rules, learner baseline and environment check
09:15 – 10:45	Session 1	The telecom business model and network anatomy from 2G to 5G
10:45 – 11:15	Break	—
11:15 – 13:00	Session 2	The telecom data landscape and KPI families; telecom data exploration lab

Ground rules

- Questions interrupt. Do not save them for the end.
- Nothing here is graded by attendance; it is graded by what you build.
- Lab work started in session is carried forward before the next day.
- Everything we use is free. No licence, no card, ever.
- One artifact folder per learner, kept for all 10 days.

4 hours a day means the room is for building. Reading and reruns happen between sessions.

Advanced AI & ML for Telecom, at a glance

10

days of instruction

4

hours per day

40

total contact hours

30

maximum cohort size

8

modules

4

ML models you train yourself

1

live application you publish

₹0

software cost, end to end

Audience: final-year engineering students with an existing foundation in AI and machine learning. No prior telecom knowledge is assumed.

Learners build. Learners do not watch.

What every one of you walks out having personally constructed

1

An automated telecom data pipeline

Four raw operator sources joined, cleaned, scheduled and logged

2

Four trained machine learning models

ARPU regression, churn classification, segmentation, anomaly detection

3

A published, live prediction app

Reachable on a public URL, not a notebook on your laptop

4

A grounded GenAI assistant

Retrieval over telecom documents, with guardrails

5

Live KPI dashboards and alerts

Grafana monitoring plus a management BI view

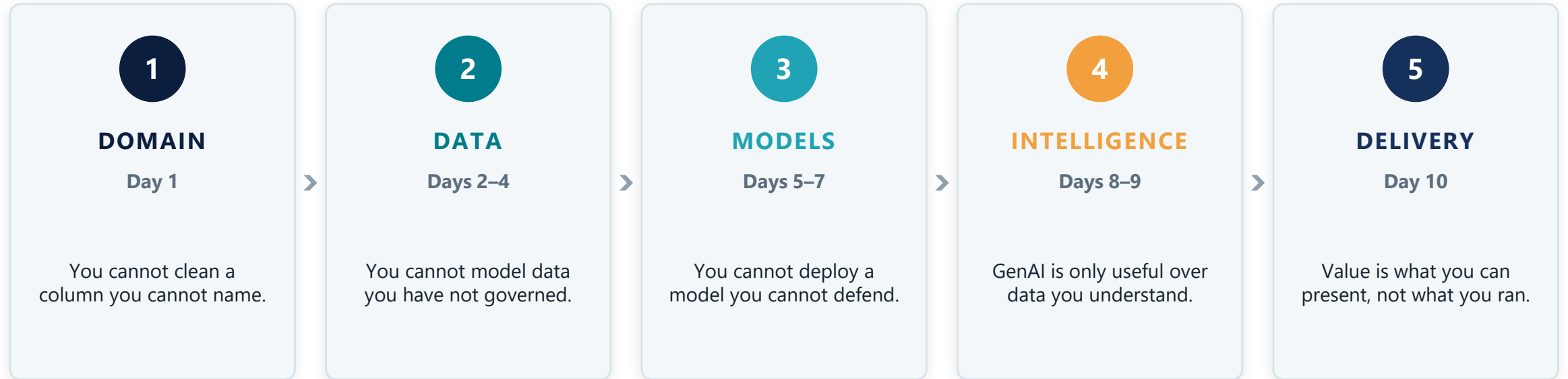
6

A public portfolio

The entire body of work, versioned and presentable at interview

WHY THE ORDER IS WHAT IT IS

Domain → Data → Models → Deployment



Generative AI arrives on Day 8 deliberately - an assistant over telecom knowledge is only meaningful once you know the KPIs and the models underneath it.

Module-wise roll-out 1 of 2

01

Telecom & 5G Foundations for AI

Day 1

Outcome

Read a telecom dataset for meaning and trace every KPI back to its source

02

Telecom Data, SQL, Python & ETL

Days 2 & 3

Outcome

Build a governed master dataset from four raw operator sources

03

Data Engineering, Scale & Governance

Day 4

Outcome

Automate that pipeline at volume, under privacy controls

04

Machine Learning for Telecom

Days 4 & 5

Outcome

Train, tune, validate and explain predictive and unsupervised models

Module-wise roll-out 2 of 2

05

Forecasting & Real-Time Analytics

Day 7

Outcome

Forecast a KPI with uncertainty and monitor it live in Grafana

06

GenAI & Agentic AI in Telecom

Day 8

Outcome

Build a grounded AI assistant and an agentic operations workflow

07

MLOps, BI & Prescriptive Analytics

Day 9

Outcome

Deploy, monitor, and turn a model into a costed recommendation

08

Capstone, Evaluation & Career Launch

Day 10

Outcome

Deliver and defend a complete solution, and evidence it publicly

Ten days, one continuous build



Each day consumes the previous day's output. Miss the build on Day 3 and Day 5 has nothing to train on — which is why between-session work is not optional.

Every tool is free. No licence, no subscription, no card.



MySQL Server + Workbench

Database practice · Days 2–4



Miniforge (Python + Jupyter)

Development environment · throughout



Pandas, scikit-learn, Matplotlib

Wrangling, modelling, plotting



KNIME Analytics Platform

Visual ETL and scheduling · Day 4



Grafana OSS

Live KPI monitoring and alerts · Day 7



Google AI Studio (free tier)

Gemini API key for GenAI · Day 8



sentence-transformers + ChromaDB

Embeddings and vector store · Day 8



Streamlit + Hugging Face Spaces

Publishing your live app · Day 9



Git + GitHub account

Version control and portfolio · Day 10

A Google account and a Hugging Face account are needed later in the programme — both are free and take an email address only.

You are measured on what you build

Daily

Polls & open Q&A

Ungraded. A quick pulse check at the start of each session to see what landed and what needs a second pass.

Days 4, 6 & 9

Mini-capstone checkpoints

Three graded checkpoints: the governed pipeline, the tuned model set, and the deployed application.

Day 10

Final capstone

A complete solution, presented and defended, assessed by the trainer against a published rubric.

The rubric is published in advance. Nothing about the assessment is a surprise.

MODULE 1 • DAY 1

Telecom & 5G Foundations for AI

The business, the network, the data, and the KPIs — before a single model is trained.

Five things, in this order

1

Network anatomy, 2G to 5G

Radio access, transport, packet core, and the OSS/BSS layer that runs and bills it

2

5G architecture and spectrum

Service-based design, slicing, edge compute, and why band choice changes everything

3

The telecom business model

How an operator makes money, loses it, and why churn is the number everyone watches

4

The telecom data landscape

Where CDRs, PM counters, alarms, tickets, billing, CRM and IoT telemetry are born

5

KPI families and their meaning

Accessibility, retainability, mobility, integrity, availability - plus ARPU, churn, downtime, CSAT

Anatomy of a mobile network

Four layers. Every telecom dataset you will touch is produced by one of them.

RAN

Radio Access Network

Antennas, radios and basebands at the cell site. Owns coverage, capacity and the air interface.

DATA IT PRODUCES

PM counters, alarms, cell-level KPIs

Transport

Backhaul & fronthaul

Fibre and microwave links carrying traffic from sites to the core. Owns latency and link availability.

DATA IT PRODUCES

Link utilisation, latency, outages

Core

Packet & signalling core

Authentication, session management, mobility, routing to the internet and to other operators.

DATA IT PRODUCES

Session records, signalling logs, CDRs

OSS / BSS

Operations & business support

Runs the network and runs the business: provisioning, assurance, charging, billing, CRM, care.

DATA IT PRODUCES

Billing, CRM, tickets, orders

Subscriber device → RAN → Transport → Core → internet, and OSS/BSS wrapped around all of it.

Digital Disruption – ***Happened***, is Happening and is bound to Accelerate

1994



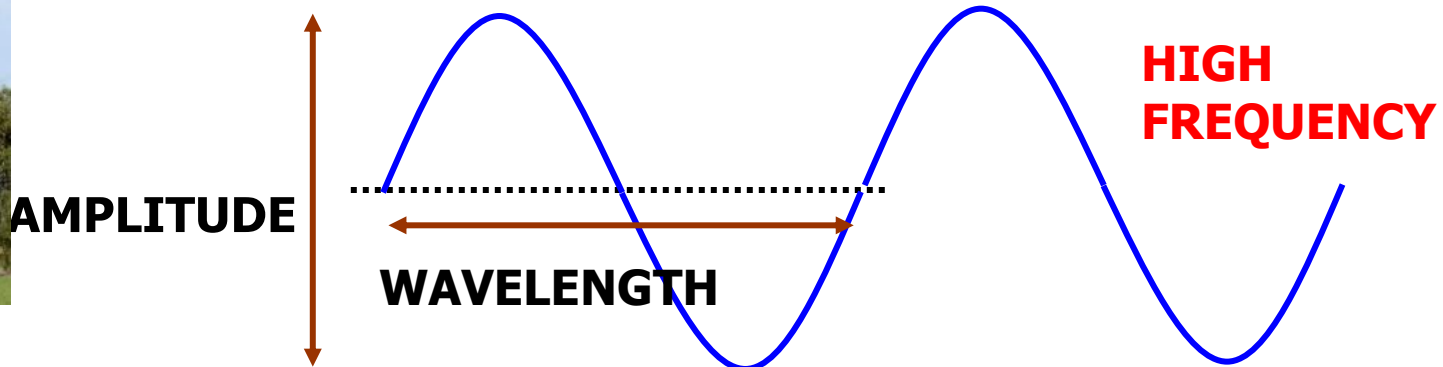
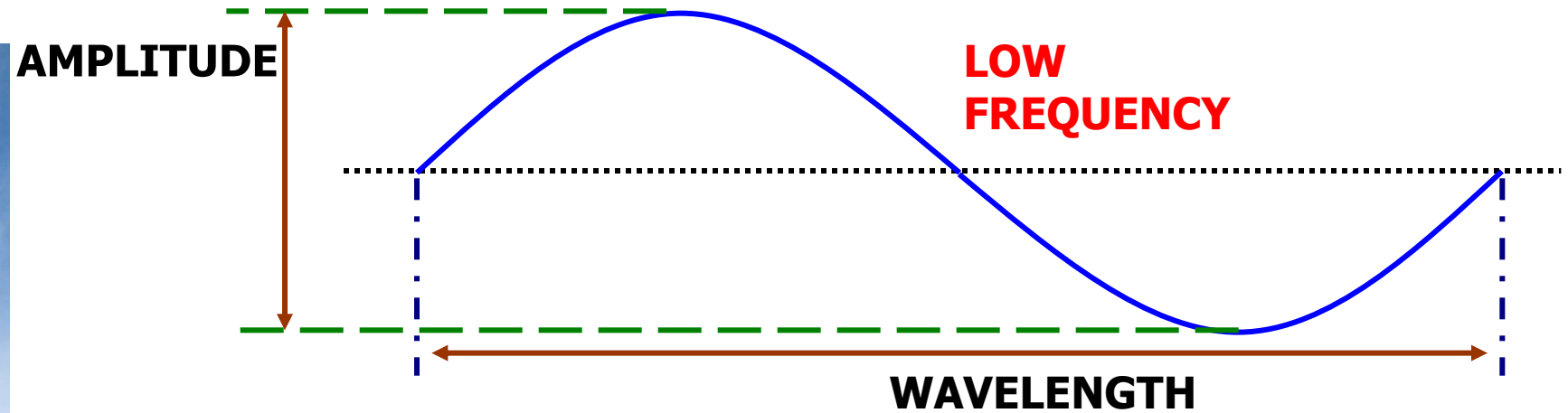
2014



2025

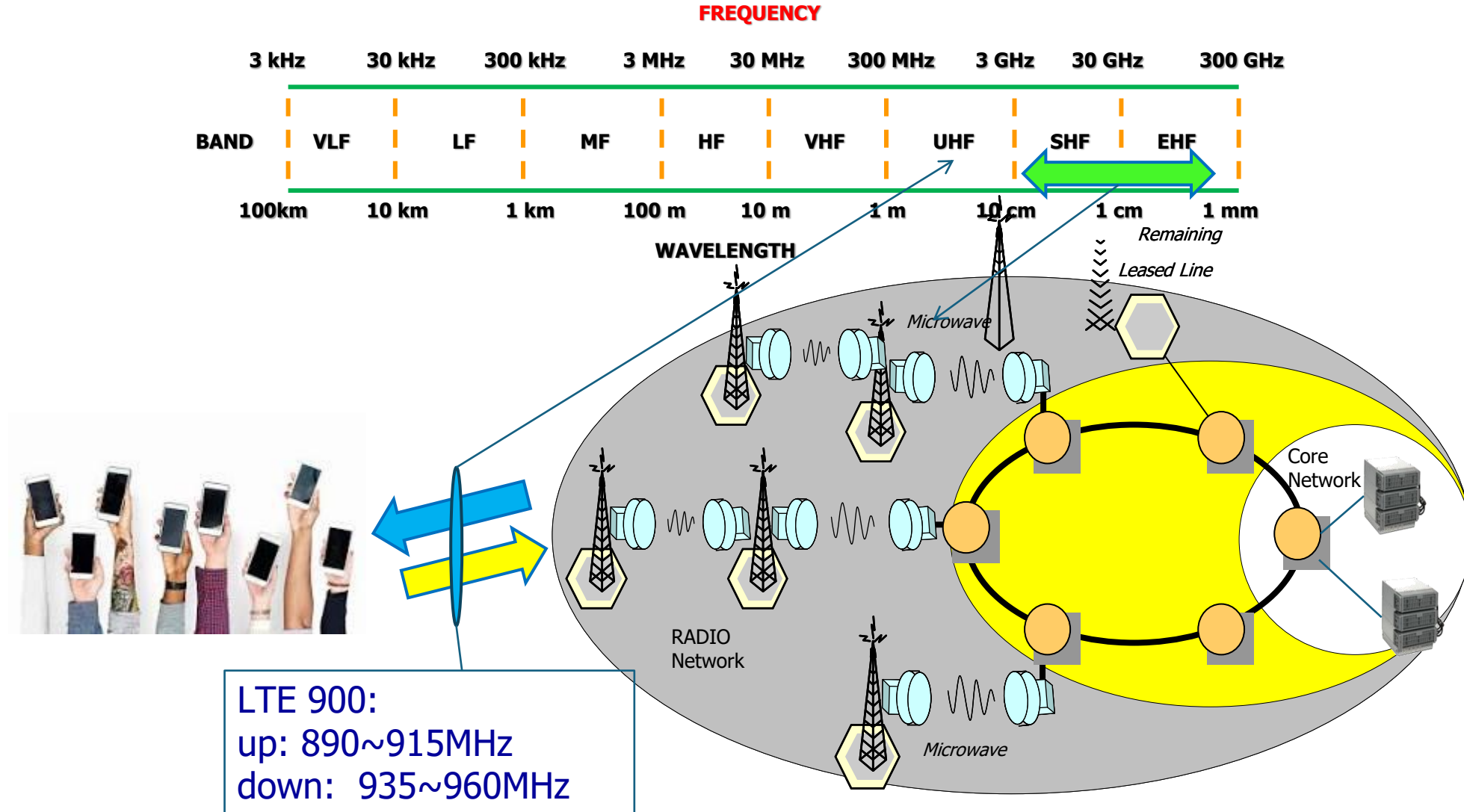


CONNECTIVITY - Frequency and Wavelength



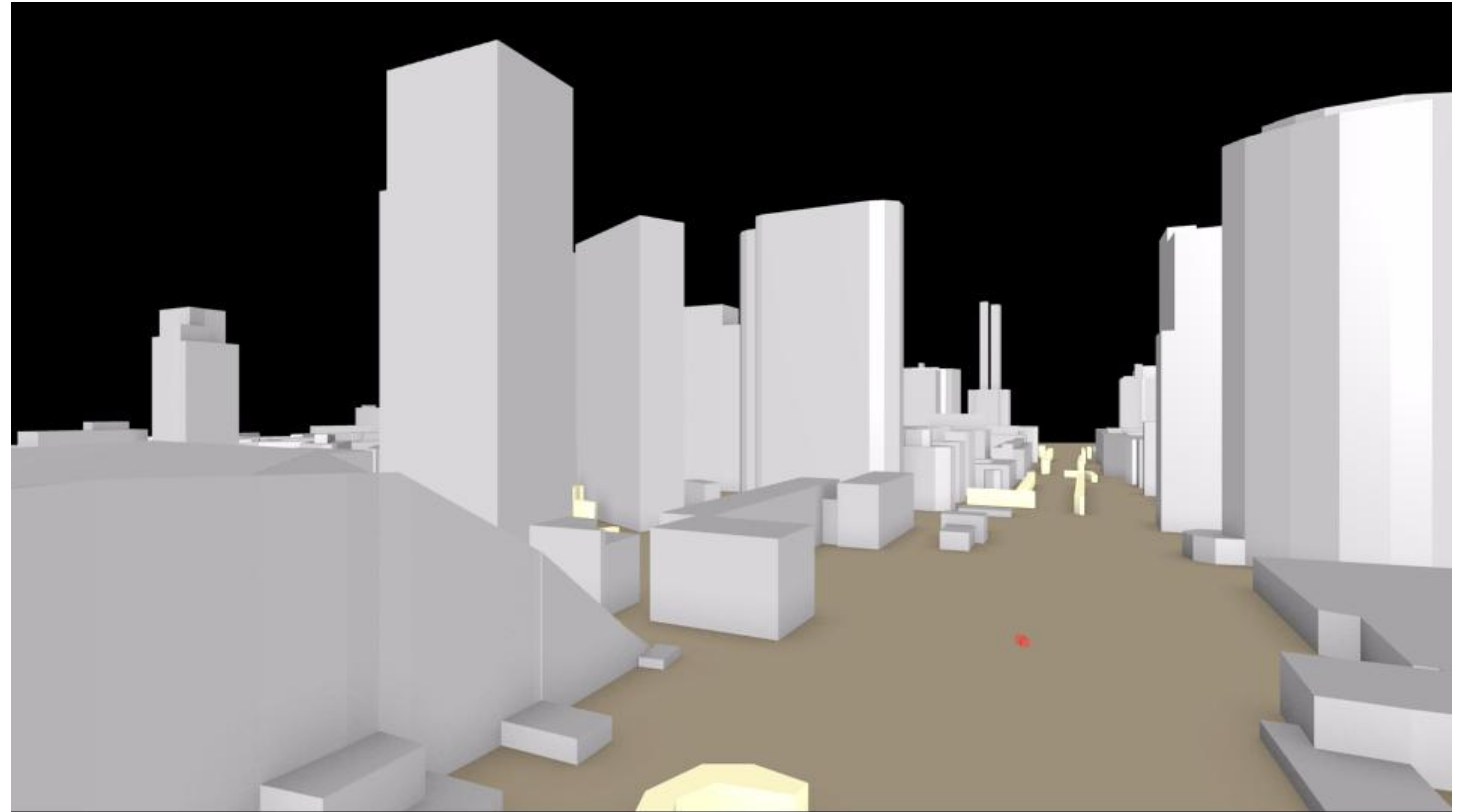
Wavelength decreases and frequency increases

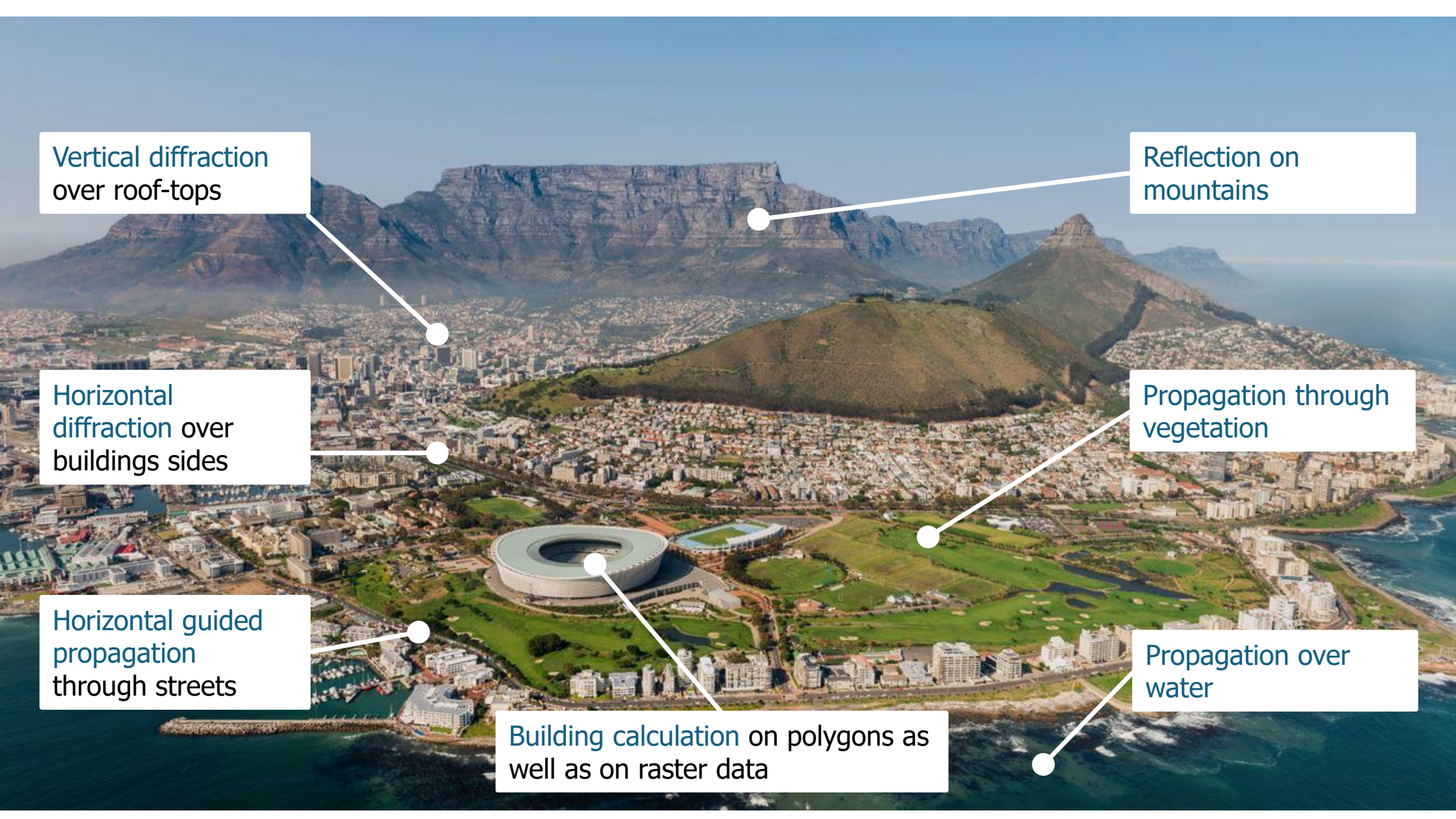
Radio Access Network



ANTENNA TRANSMIT

- What is seen is the direct signal being transmitted and leaving the antenna in every direction at the same time, effectively taking the shape of a sphere that grows with time.
- As the sphere passes through the camera, the camera is turned back to reveal the great Arch and the sphere reaching it.





Vertical diffraction
over roof-tops

Reflection on
mountains

Horizontal
diffraction over
buildings sides

Propagation through
vegetation

Horizontal guided
propagation
through streets

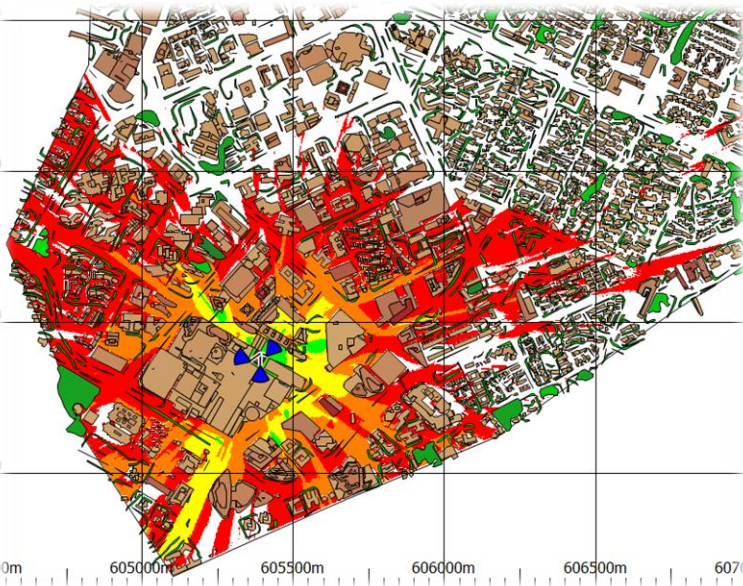
Propagation over
water

Building calculation on polygons as
well as on raster data

5G Coverage – 700 Mhz vs 3.5 GHz

- Outdoor coverage plot 700 MHz (low band) vs 3.5 GHz (mid band)

700 MHz

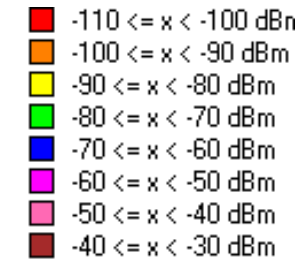


Total 2D View area (km²):	7.261
Covered area(km²):	1.065
Covered area(%):	14.669%

3.5 GHz



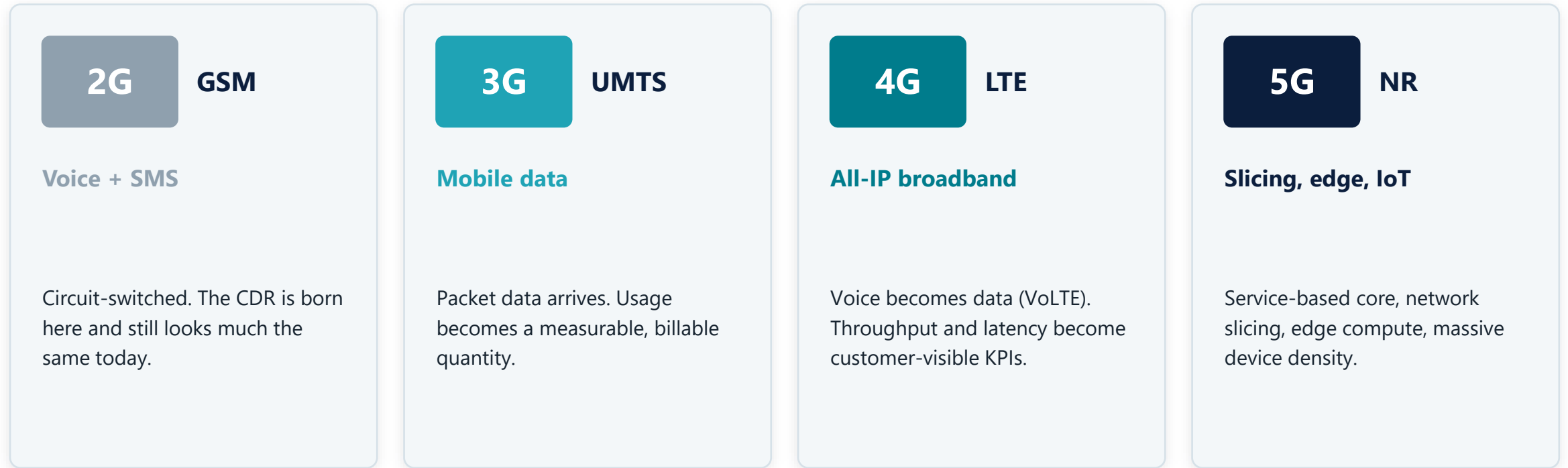
Total 2D View area (km²):	7.261
Covered area(km²):	0.070
Covered area(%):	0.961%



~15 times lower
foot print in terms of
covered area using
28 GHz band!

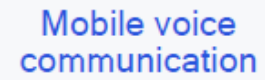
* mmWave 30 GHz to 300 GHz => wavelength 10 mm –to 1 mm

2G to 5G - what actually changed



An Indian operator runs all four generations simultaneously. Your data will contain 2G voice records, 4G data sessions and 5G counters in the same month — and they do not mean the same thing.

Evolution of cellular technologies ~ 10 years



1G

1980s

Analog voice
AMPS, NMT,
TACS

Efficient voice to reach billions

2G

1990s

Digital voice

D-AMPS, GSM, IS-95 (CDMA)

Focus shifts
to mobile data

3G

2000s

Wireless Internet
CDMA2000/EV-DO
WCDMA/HSPA+,

Mobile broadband and emerging expansion

4G

2010s

Mobile broadband
LTE, LTE Advanced,
Gigabit LTE

A unified future-proof platform

5G

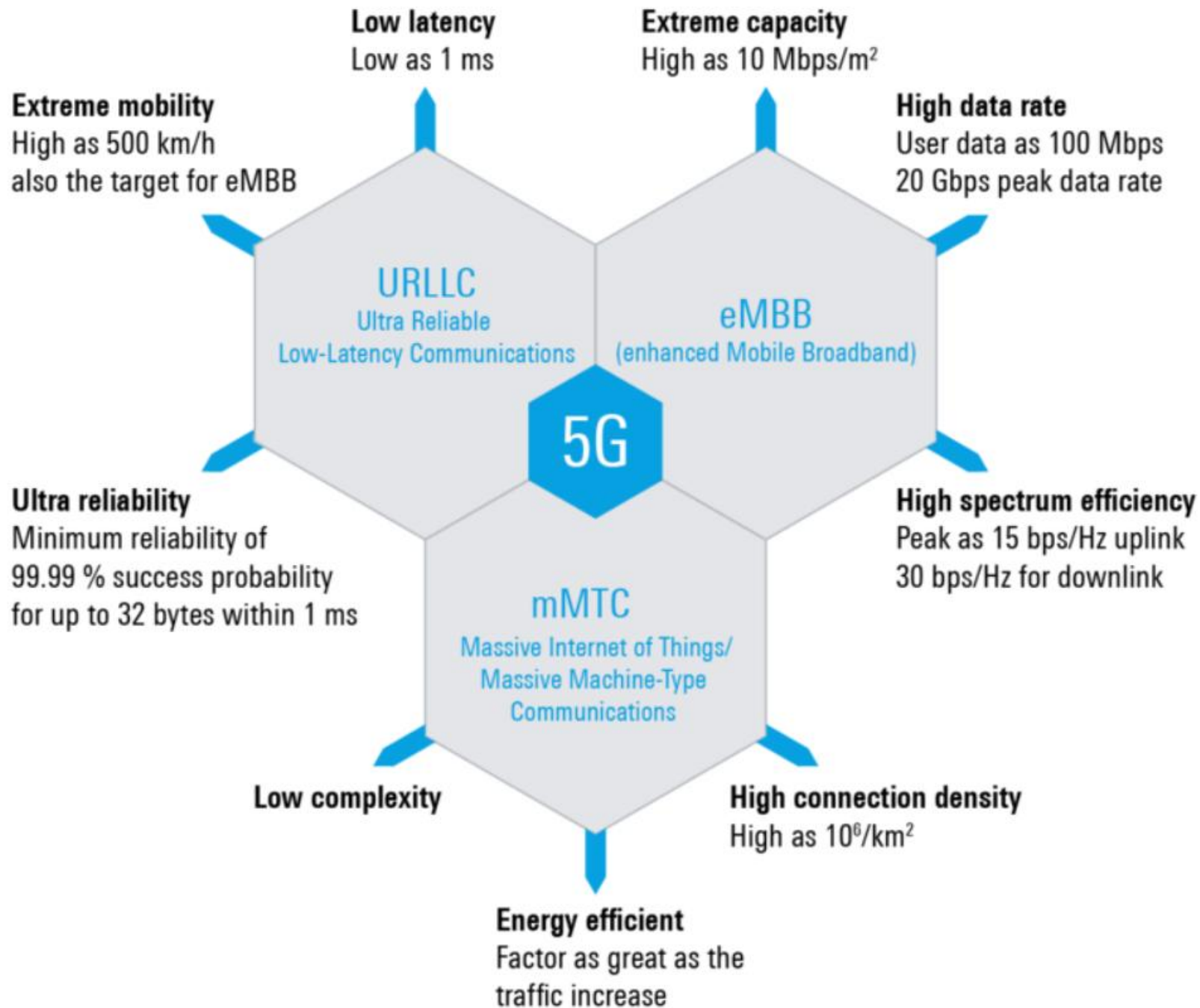
2020s

Wireless Edge

5G NR^T

5G New *Thinking*

Its fundamental shift in thinking and 5G aims to make it possible. Revolutionary & Evolutionary mix bag
IT & Telecom in true sense merging to create digital society



5G: three service classes, one network

eMBB

Enhanced Mobile Broadband

Peak throughput for video, AR and fixed wireless access.

Optimises: data rate

URLLC

Ultra-Reliable Low-Latency

Millisecond latency and near-total reliability for automation, robotics, remote control.

Optimises: latency & reliability

mMTC

Massive Machine-Type Comms

Up to a million low-power devices per square kilometre — meters, sensors, trackers.

Optimises: device density



Service-based architecture

The core is a set of network functions calling each other over APIs — software, not boxes.



Network slicing

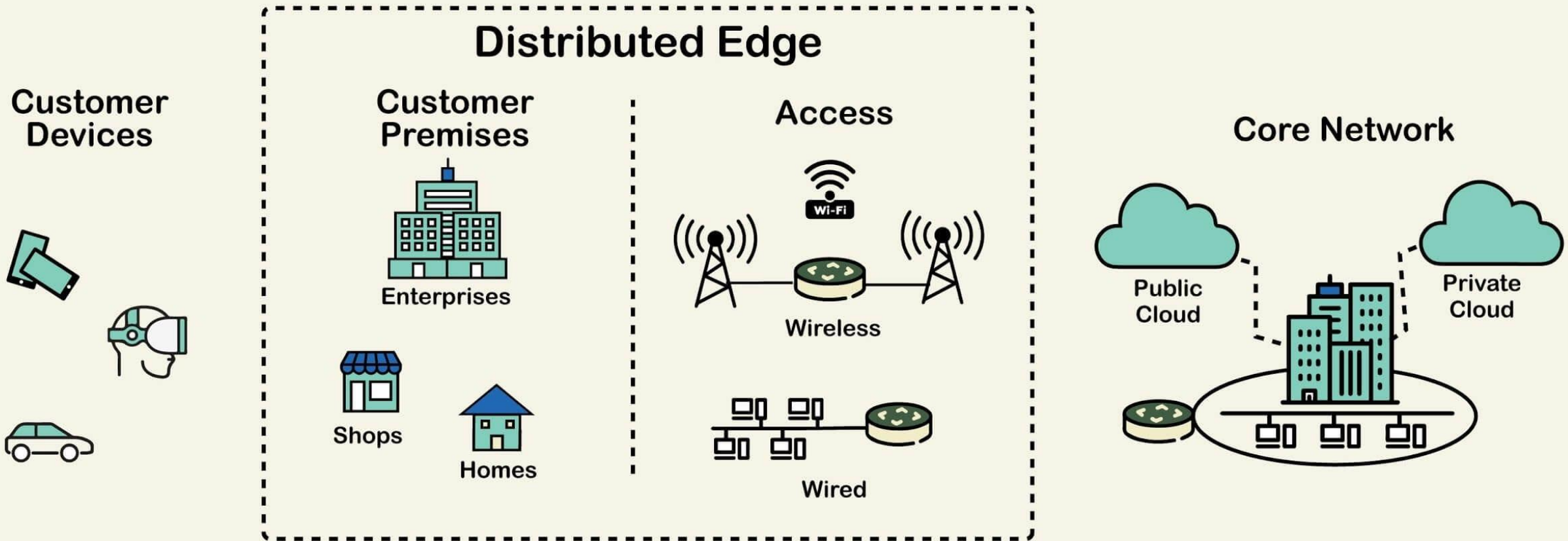
One physical network, many logical networks, each with its own performance contract.



Edge compute

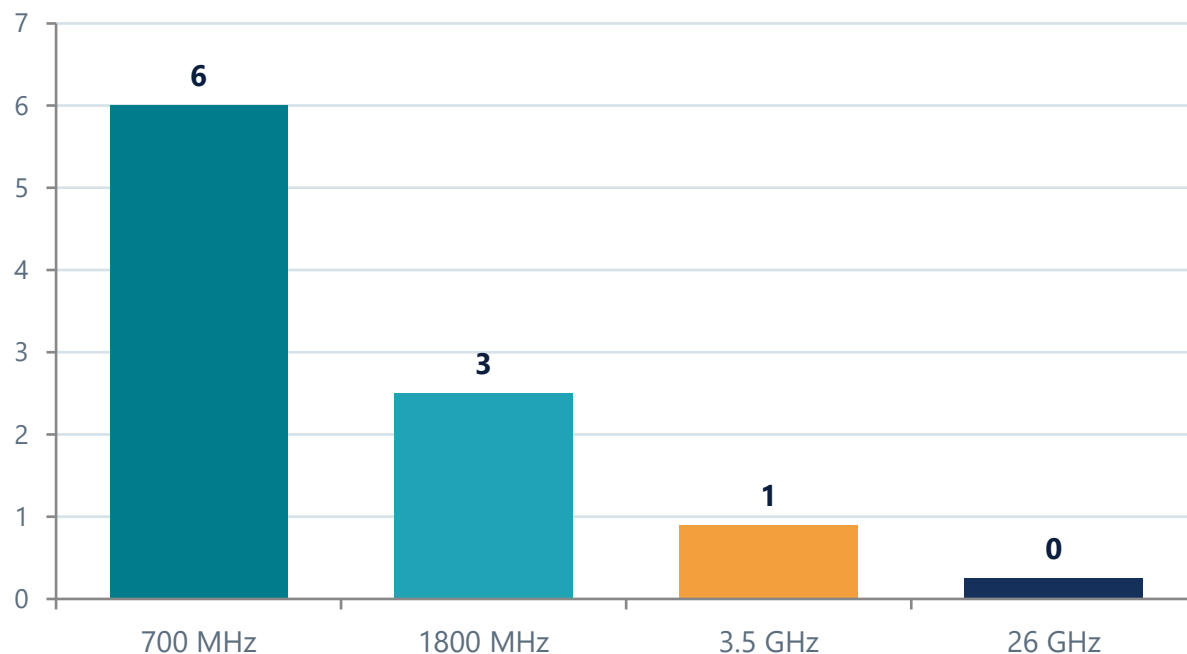
Processing moved close to the cell site, because physics will not let latency be negotiated.

High Level Network Architecture



Spectrum: coverage and capacity pull in opposite directions

Lower band = wider cell. Higher band = more capacity, less reach.



Low band · 700–900 MHz

Travels far, penetrates walls. Used for rural coverage and indoor depth. Little capacity per cell.

Mid band · 1800 MHz – 3.5 GHz

The workhorse. The compromise every urban 4G and 5G network is actually built on.

High band · mmWave 26 GHz

Enormous capacity over a few hundred metres, blocked by walls, rain and hands. Stadiums and hotspots.

This is why a rural region shows lower throughput and higher drop rates than a metro before anyone blames the operator's engineering.

Now the regional variation in your data has a cause

OBSERVED IN THE DATA

A region shows throughput 30% below the others

PHYSICAL EXPLANATION



Its coverage is carried mostly by low band. It is a spectrum allocation, not a fault.

OBSERVED IN THE DATA

Drop rate spikes every evening at 20:00–22:00

PHYSICAL EXPLANATION



Cell congestion at busy hour. A capacity problem, not a quality-of-equipment problem.

OBSERVED IN THE DATA

ARPU is high but complaint volume is also high

PHYSICAL EXPLANATION



Dense urban, heavy users, congested sites. Retention risk hides inside your best revenue.

OBSERVED IN THE DATA

Two regions have identical ARPU and different churn

PHYSICAL EXPLANATION



Competitive intensity differs. The model needs a feature that captures it.

A model that does not encode the cause will keep rediscovering the symptom.

5G Phones



5G Wearable



5G Silicon | Chip | Device Manufacturers



5G Tower | DAS | SmallCells



Cloud Service Providers



5G Management & Orchestration



5G Testing



5G Radio Access | 5G NR



5G vEPC | 5G Core



5G vIMS



5G Transport & SDN



5G Network Slicing



5G OSS | BSS



5G Security



5G Network Rollouts by Network Operators Live Deployments as of Aug 2020



5G ECOSYSTEM August 2020



5G Assurance



SASE



SD-WAN



5G System Integrators



5G Virtual Infra Manager (VIM)



5G Open Source Projects & Labs

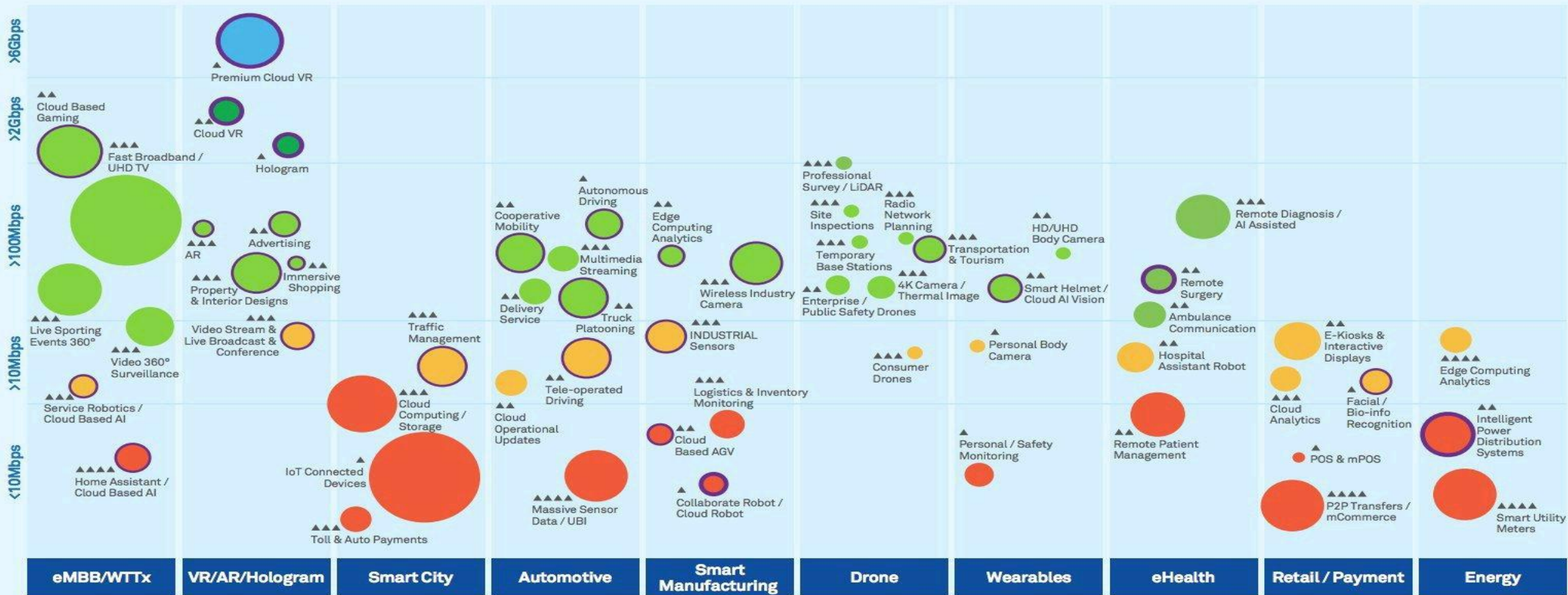


SDOs | Market Partners & Alliances



5G Applications Market Potential & Readiness Matrix

Connectivity & Value Added Services Opportunity



Index Definitions

2020 Readiness for 5G Index

1. Prototype-----▲
2. Early Adopter-----▲▲
3. Price Reduction-----▲▲▲
4. Feature Enhanced-----▲▲▲▲
5. Mature Application-----▲▲▲▲▲

2025 Market Potential Index



Latency Requirement Index

- Low latency required (<50 ms)
- Very low latency (<10 ms)

Bandwidth Requirement Index

- Low: <10 Mbps-----●
- 11 to 99 Mbps-----●
- Med: 100M to 1Gbps-----●
- 2 to 5 Gbps-----●
- High: 6 to 10 Gbps-----●

How a telecom operator makes money

$$\text{Revenue} = \text{Subscribers} \times \text{ARPU}$$

Only three levers exist: add subscribers, keep subscribers, or raise what each one spends. Every telecom analytics project in existence serves one of those three.

1 Acquire

Cost per acquisition is high — handset subsidy, channel commission, onboarding. A subscriber who leaves in month three never repaid it.

2 Retain

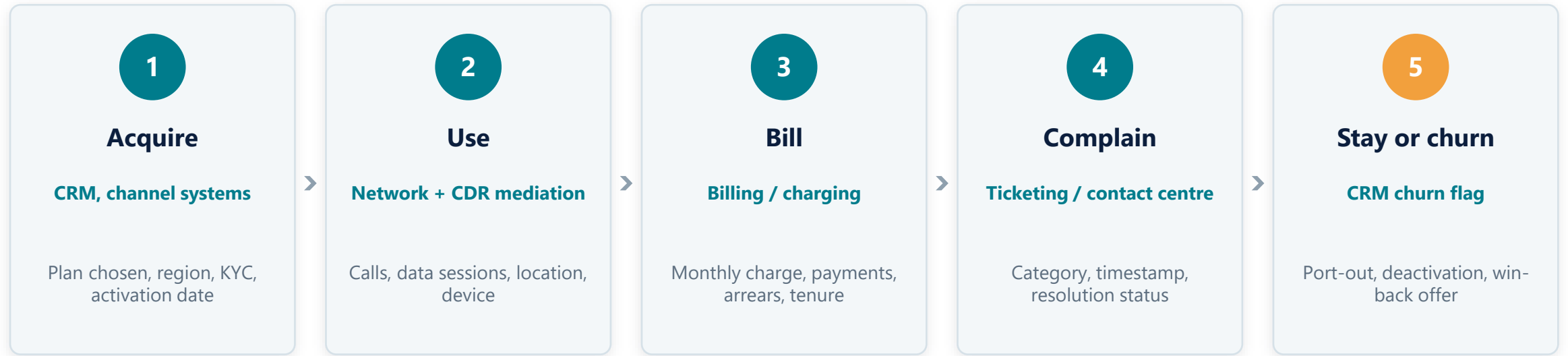
Churn is the most expensive number on the sheet. A 1-point monthly churn reduction on 10 million subscribers is worth more than most price rises.

3 Grow ARPU

Upsell to higher data tiers, bundle content, convert prepaid to postpaid, sell enterprise and IoT connectivity.

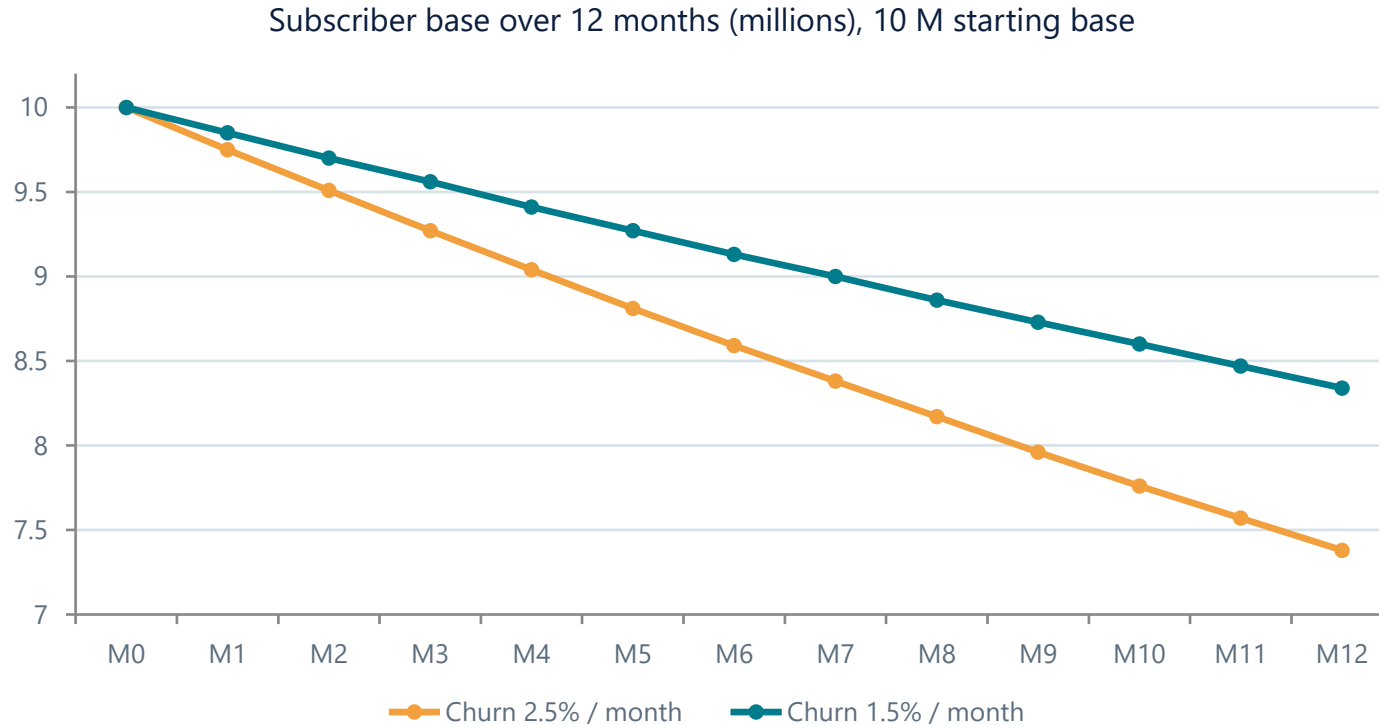
The subscriber lifecycle leaves a data trail

Each stage writes to a different system — and that is exactly why your data arrives in four files, not one



On Day 2 you will join customers, usage, billing and complaints into one table. The join keys exist because the lifecycle above is one journey recorded by five different owners.

Why churn is the number everyone watches



One point of churn

- At 2.5% monthly churn the operator loses roughly 2.6 million subscribers in a year and must acquire that many just to stand still.
- At 1.5% the annual loss is about 1.7 million. The gap is close to 0.9 million paying subscribers.
- At ARPU of ₹180 per month, that gap is worth well over ₹150 crore of annualised revenue.
- No pricing campaign moves the number that far. This is why churn prediction is Day 5.

...and how an operator loses money

Churn

Subscribers port out to a rival. Usually visible in the data weeks before it happens — falling usage, rising complaints, unpaid bills.

Revenue leakage

Usage that is delivered but never billed: mediation gaps, rating errors, unbilled roaming, misconfigured plans.

Fraud

SIM box bypass, subscription fraud, IRSF. Shows up as statistical anomalies, never as a labelled column.

Network cost

Capex on sites that carry little traffic; opex on energy, backhaul and field maintenance for underused capacity.

Poor quality

Dropped calls, congestion and outages drive complaints, then CSAT down, then churn up. It is one causal chain.

Bad targeting

Discounts pushed to subscribers who were never going to leave. Retention spend without retention gain.

Every one of these six becomes a modelling problem later in the programme. None of them can be framed correctly without the domain understanding you are building today.

Where telecom data is born

CDR

Call Detail Records

Mediation from the core

Per call / session

PM

Performance counters

RAN and core elements

Per cell, 15 min or hourly

FM

Alarms & faults

Network management system

Event-driven

TT

Trouble tickets

Assurance / field ops

Per incident

BIL

Billing & charging

BSS billing platform

Per bill cycle

CRM

Customer & care

CRM, contact centre

Per interaction

IoT

Device telemetry

IoT platform, 5G mMTC

Continuous, high volume

Ask of every source

- Who owns it?
- What is its grain?
- How late does it arrive?
- Can it be trusted?

The scale you are being trained for

50 M+

CDRs generated per day by a mid-sized operator

500 GB

raw usage and counter data landing every day

10 K

analyst and dashboard queries per hour at peak

15 min

the granularity at which most PM counters report

What this means for how you work

- A pandas read_csv of the whole table is not a strategy — you sample, filter and aggregate at source.
- Pipelines must be scheduled, idempotent and logged, because nobody re-runs 500 GB by hand.
- Late-arriving data is normal; your daily report must be correct anyway.
- Days 3 and 4 are entirely about making this survivable. Today is about knowing what is coming.

The four commercial KPIs this programme is built around

ARPU

Average Revenue Per User

Total revenue ÷ average subscribers, per month

Moves slowly. A 2-point move is a board conversation.

Churn rate

Subscriber loss rate

Disconnections ÷ opening subscriber base, per month

The single most modelled target in telecom analytics.

Downtime

Service unavailability

Hours a cell or service was unavailable in the period

The bridge between network quality and commercial pain.

CSAT

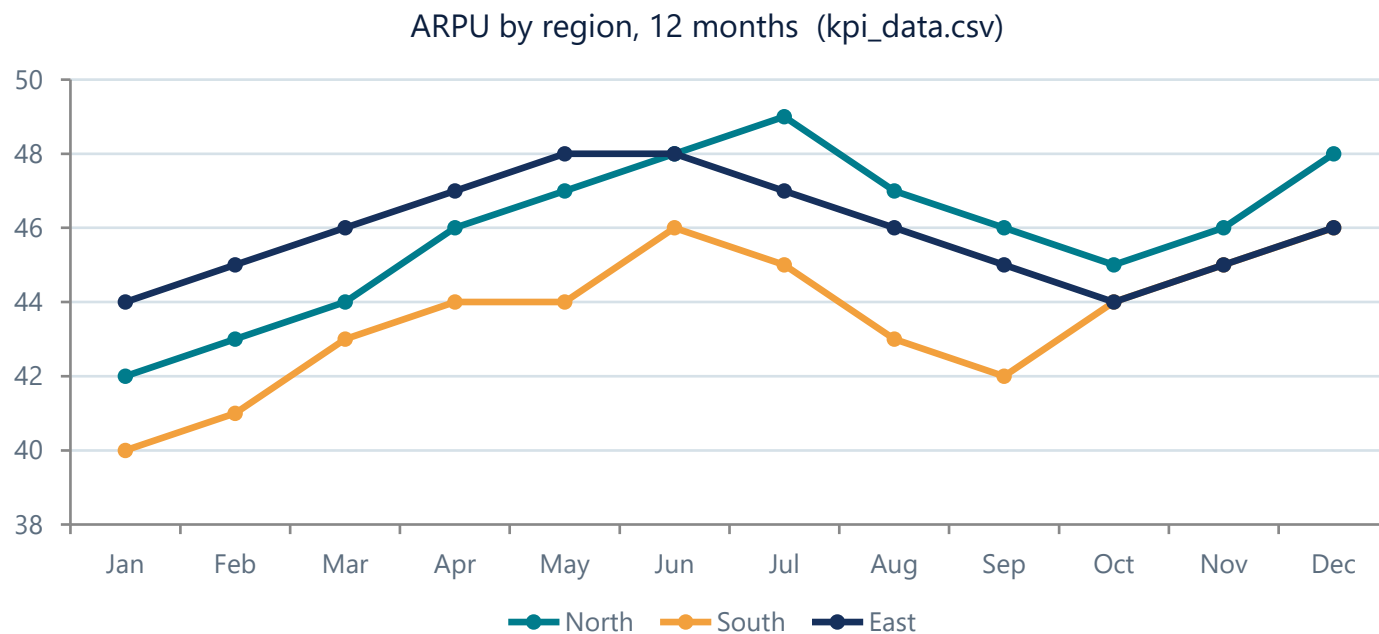
Customer satisfaction

Mean satisfaction score from surveys, 1 to 5

Lagging, sparse and biased — treat with care as a target.

These four are the columns of the dataset you are about to open — and the targets of the models you will train on Days 5, 6 and 7.

Reading the course KPI dataset



Six columns, thirty-six rows

- Grain: one row per region per month. Not per subscriber, not per cell.
- All three regions rise to a mid-year peak, then fall back — seasonality, not a campaign.
- South sits consistently below the others by 2–4 points. That gap is the interesting question.
- Downtime and churn move together; CSAT lags both by a month.

Before you model anything: what is the grain, what is the period, what is missing, and who produced it? Those four questions are the whole of today's first exercise.

From counter to decision — the chain you must be able to walk

- 1 Network element** A 4G cell in the South region counts every RRC connection attempt and every success.
- 2 Raw counter** $\text{rrc_conn_att} = 128,400 \cdot \text{rrc_conn_succ} = 121,600$, reported every 15 minutes.
- 3 KPI** Accessibility = 94.7% - below the 98% target for eight consecutive days.
- 4 Business signal** Complaints from that region rise; CSAT drops 0.3; churn among high-ARPU users ticks up.
- 5 Decision** Add capacity at four sites, or accept the churn cost. Someone must be able to justify either.

A senior analyst can walk this chain in both directions — from a number on a dashboard back to the element that produced it.



END OF DAY 1

Domain first. Everything else stands on it.

Advanced AI & ML -Telecom Data Analytics · Nokia Initiative

Ishan Marwah · MSM Technologies

